MIL-C-14841A(MU)
12 October 1973
SUPERSEDING
MIL-C-14841(MU)
20 July 1969

MILITARY SPECIFICATION
CARTRIDGE, DELAY: M231, ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Delay: M231, for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-60539	- Coating, Anodic, Conventional, for Aluminum and Aluminum Alloys

STANDARDS

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MS-28741	- Hose Assembly, Detachable End-fitting, Medium Pressure

DRAWINGS

Armament Command

D10536005	- Cartridge, Delay, M231, Assembly
C8799620	- Box, Packing, Ammunition for Cartridge, Delay, M70, M93, M113, M155 and M231.

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:53Z
Check the source to verify that this is the current version before use.

C8799621 - Carton, Packing, Ammunition for Cartridge, Delay, M70, M93, M113, M135, or M231

IEL-10536005 - Inspection Equipment List for Cartridge Delay, M231

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions shoudl be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N. Y. 10018).

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.-Protective finish shall be in accordance with MIL-C-60539 and shall apply to all surfaces of the assembled cartridge regardless of configuration.

3.3 Vibration.-The cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant charge.

3.4 Primer sensitivity.-The primer shall fire when drop tested at an energy level of 26 inch ounces in a sectioned cartridge case, void of delay charge, igniter and propellant charge.

3.5 Functional.

3.5.1 The cartridge shall not fail to fire.

3.5.2 Ballistics.-The cartridge when tested in an M114 initiator shall produce the pressures and delay times at the temperature specified in Table I. Pressure values shall be recorded at the end of 15 feet of aircraft hose, size no. 4, in accordance with MS-28741 and a pressure chamber with a volume of 1 cubic inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:53Z
Check the source to verify that this is the current version before use.

TABLE I

Temperature	Pressure (PSI) Min.	Max.	Delay (Seconds) (See 6.2)
-65°F to -70°F	500	1250	0.93 Max.
65°F to 75°F	600	1500	0.72 ± 0.12
200°F to 205°F	700	1700	0.57 Min.

1/ Pressure values shall be produced in a one cubic inch pressure station at the end of 15 feet of hose (size no. 4) in accordance with MS 28741.

3.6 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.-The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr.-The part shall not have a burr which would interfere with the assembly or function of the part or be injurious to personnel handling the item.

3.6.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:53Z
Check the source to verify that this is the current version before use.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 50 cartridges for primer sensitivity, 5 cartridges for sealant tests (to be selected from primer sensitivity sample), 30 cartridges for ballistic tests, 12 for vibration tests (to be selected from ballistics sample) and 10 sets of unassembled components for dimensional and visual materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.-After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with Table I. The sample for firing shall consist of the following:

TABLE II

Temperature	Not Vibrated	Vibrated
-65° to -70°F	6	4
65° to 75°F	6	4
200° to 205°F	6	4

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105

4.3.1.2 Size of lot.-Unless otherwise specified by the procuring agency, an ammunition lot shall consist of an identifiable group of metal parts assemblies consisting of:

a. Components produced by a homogeneous process.
b. Delay charge from not more than one lot.
c. Primers from not more than one lot.
d. Igniters from not more than one lot.
e. Propellant from not more than one lot.

Source: https://assist.dla.mil -- Downloaded: 202610-18T04:53Z
Check the source to verify that this is the current version before use.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects using the sampling plan and associated criteria in Tables III and IV. For an individual defect (single characteristic) the criteria in Table III shall be used. For total defects within a major or minor class, the criteria in Table IV shall be used. All non-conforming material shall be rejected. Four power (4X) magnification may be used to perform visual examination for such defects as burr and foreign matter. The roughness comparison specimen prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination.

TABLE III

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 to 300	100% Examination	-	-	-	-
301 to 1300	110	1	2	2	3
1301 to 3200	150	2	3	3	4
3201 to 8000	225	3	4	4	5
8001 to 22000	300	4	5	5	6

TABLE IV

	Total allowable defects per sample size			
Sample Size for Major Defects - - - - - - - -	110	150	225	300
Number of Characteristics in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Size for Minor Defects - - - - - - - -	110	150	225	300
Number of Characteristics in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

Source: https://assist.dla.mil -- Downloaded: 202610-18T04:53Z
Check the source to verify that this is the current version before use.

Defect	Method of Inspection
CRITICAL	
1. Mixed delay elements	Visual
2. Flash hole missing or obstructed	Visual
3. Primer uncrimped or seal missing	Visual
4. Marking missing, incorrect or illegible	Visual
5. Delay charge not visible through flash hole	Visual
6. Metal defective	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
103. Protective finish inadequate or defective	Visual
MINOR	
None defined	

4.3.2.2 Disc, Sealing Dwg. C8594077, covering a detail of Dwg. D10536005.

CRITICAL	
1. Metal defective	Visual
2. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Diameter, min.	Gage
102. Thickness	Gage
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter, max.	Visual
202. Varnish missing or inadequate	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:53Z
Check the source to verify that this is the current version before use.

4.3.2.3 Case, Cartridge, Dwg. C8595202, covering a detail of Dwg. D10536005.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Straightness of base	Gage
103. Diameter of body, min.	Gage
104. Diameter of head, min.	Gage
105. Location of datum diameter	Gage
106. Thickness of base	Gage
107. Thickness of wall	Gage
108. Protective finish inadequate or defective	Visual
109. Burr or sharp corner	Visual
110. Foreign matter	Visual
MINOR	
201. Total length	Gage
202. Diameter of head, max.	Gage
203. Radii or chamfer incorrect or missing	Visual
204. Diameter of body, max.	Gage
205. Surface finish improper	Visual

4.3.2.4 Cartridge, Delay, M231 Assembly Dwg. D10536005.

CRITICAL	
1. Any part missing	X-ray & Visual 1/
2. Improper assembly	X-ray & Visual 1/
3. Mixed delay element or propellant	X-ray & Visual 1/
4. Metal defective	Visual
5. Crimp missing or incorrect	Visual
6. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Diameter of body, max.	Gage
102. Diameter of head, max.	Gage
103. Location of datum diameter	Gage
104. Cartridge profile failure	Gage
105. O-ring split, cracked or cut	Visual
106. Protective finish inadequate or defective	Visual
107. Burr	Visual
108. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:53Z
Check the source to verify that this is the current version before use.

Defect	Method of Inspection
MINOR	
201. Total length	Gage
202. Case damaged	Visual
203. Varnish missing or inadequate	Visual

1/ The assembled cartridge shall be x-rayed with the long axis of the Cartridge perpindicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring agency.

4.3.2.5 Carton, Packing, Dwg. C8799621.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damaged through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.6 Box, Packing, Dwg. C8799620.

CRITICAL	
None defined	
MAJOR	
None defined	

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:53Z
Check the source to verify that this is the current version before use.

4.3.2.6 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual & Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness.-Ten units from each heat treated lot shall be tested to determine compliance with the drawing requirements. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.2 Protective finish.-Unless otherwise specified, sampling plans and procedures in determination of acceptability of coated parts and articles submitted by a supplier shall be in accordance with provisions set forth in Table III and IV and MIL-C-60539.

4.3.3.3 Vibration.-Twelve of the assembled cartridges of the initial production sample or 12 assembled cartridges from a subsequent lot (when a change has been made in the lot of primers, propellant or delay elements) shall be subjected to vibration as directed in 3.3. After vibration the 12 cartridges shall be x-rayed in accordance with note 1/ of 4.3.2.4. Failure of any cartridge to comply with the requirements of 3.3 shall be cause for rejection of the lot. When there is a change in the method of charging the cartridge or in the lot of primers, 12 additional cartridges shall be selected and tested as indicated herein.

4.3.3.4 Sealant application.-Five cartridges from each production lot (from primer sensitivity sample), shall be tested to determine compliance with the drawing requirements. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.5 Airtightness.-Each cartridge shall be subjected to an airtightness test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:53Z
Check the source to verify that this is the current version before use.

4.3.3.6 Primer sensitivity.-A sample of 50 assembled cartridges from the initial production sample and 25 assembled cartridges from each subsequent production lot shall be tested for sensitivity. Failure of one cartridge to comply with the requirements of 3.4 shall be cause for rejection of the lot.

4.3.3.7 Functional.-The following test shall be performed by the Government.

4.3.3.7.1 Ballistics.-Thirty test cartridges from each production lot shall be grouped and temperature conditioned as prescribed in 4.4.7.1.1 and Table V. Failure of any cartridge to comply with the requirements of 3.5.2 shall be cause for rejection of the lot.

TABLE V

Temperature	Production Lot
-65° to -70°F	10
65° to 75°F	10
200° to 205°F	10

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL 10536005.

4.4 Test methods and procedures.

4.4.1 Hardness.-A hardness testing machine shall be used to determine the hardness values specified on the applicable drawings.

4.4.2 Protective finish.-The method and procedure for performing the protective finish tests shall be in accordance with MIL-C-60539. The case shall be sectioned longitudinally to better expose the critical surfaces to the salt spray.

4.4.3 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.4 Sealant application.-The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied (360°) of the joining surfaces at the junction of the sealing disc and case sidewall and that the propellant cavity is free of sealant.

4.4.5 Airtightness.-This test shall be performed by using the equipment listed on IEL 10536005.

Source: https://assist.dla.mil -- Downloaded: 202610-18T04:53Z
Check the source to verify that this is the current version before use.

4.4.6 Primer sensitivity.-Remove and discard the charged delay body and all powder charges from the test cartridges by sectioning the cartridge case around its periphery, one inch, maximum, above the base. Drop test 50 sectioned cartridges from the initial production sample and 25 sectioned cartridges from each production lot (all void of charged delay body and powder charges) at 13 inches, using a steel ball 15/16 inches in diameter, weighing 1.94 ounces plus or minus 0.02 ounces, in a primer sensitivity machine with a cartridge holder approved by the procuring activity.

4.4.7 Functional.

4.4.7.1 Ballistics.-The test sample shall be rigidly supported in the fixture and shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded the sample units shall be reconditioned in accordance with 4.4.7.1.1.

4.4.7.1.1 Temperature conditioning.-A standard inert cartridge containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Dwgs. C8799620 and C8799621, except that all marking shall reference XM231 Cartridge.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Delay time.-Delay time is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

6.3 Loaded Assembly Hazard Information:
Explosive Hazard Classification: Class 3
Storage Compatibility: Group B
Department of Transportation (DOT) Hazard Class C
DOT Marking - Small Arms Ammunition

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:53Z
Check the source to verify that this is the current version before use.

6.4 Hazard notice.-The cartridge described herein and certain of its components are flammable and/or explosive, consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1377-A561

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:53Z
Check the source to verify that this is the current version before use.